

OpenCPU SDK

FOTA API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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Sales@neoway.com

Support@neoway.com

Website: <http://www.neoway.com>

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Li Leilei

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the FOTA upgrade related APIs.

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2 Data Structure

2.1 nwy_ota_package_t

```
typedef struct {  
    uint32 total_size;        //ota package total size (byte)  
    uint32 offset;           //data will be written to device by offset  
    uint32 len;              //the length of data  
    uint8 *data;             //ota data  
} nwy_ota_package_t;
```

2.2 nwy_fota_download_mode_e

```
typedef enum{  
    NWY_FOTA_DOWNLOAD_HTTP = 0, //download base version ota package by http  
    NWY_FOTA_DOWNLOAD_FTP = 1,  //download base version ota package by ftp  
    NWY_FOTA_DOWNLOAD_MAX  
}nwy_fota_download_mode_e;
```

2.3 nwy_ota_http_param_t

```
typedef struct {  
    char *url;  
    int port;  
}nwy_ota_http_param_t;
```

2.4 nwy_ota_ftp_param_t

```
typedef struct {  
    char *ip;  
    int port;  
    char *username;  
    char *password;  
    char *filename;  
}nwy_ota_ftp_param_t;
```

2.5 nwy_ota_download_param_t

```
typedef struct {  
    nwy_ota_http_param_t  http_param;      //base version ota package download  
    nwy_ota_ftp_param_t   ftp_param;  
    nwy_fota_download_mode_e dl_mode;      // base version ota package download mode  
    //http or ftp  
} nwy_ota_download_param_t;
```

3 APIs

The interface function definition can be found in `nwy_fota_api.h` for FOTA upgrade-related operations.

3.1 `nwy_fota_write`

Format	<code>int nwy_fota_write (nwy_ota_package_t *ota_pack)</code>
Description	Writes the FOTA package.
Parameter	<code>ota_pack</code> : Contents of the FOTA package.
Return Value	Success: Size of the data successfully written. Failure: a value <0

3.1 `nwy_fota_update`

Format	<code>nwy_error_e nwy_fota_update (int mode);</code>
Description	Initiates the FOTA upgrade process.
Parameter	<code>mode</code> : 0: Upgrade after the next restart. 1: Immediately restart and start the upgrade.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: other error code

3.2 `nwy_fota_result_get`

Format	<code>nwy_error_e nwy_fota_result_get (int *result);</code>
Description	Retrieves the result of the FOTA upgrade.
Parameter	<code>result</code> : Upgrade result. 0: Failed to retrieve the upgrade result. 1: Upgrade successful. 2: Upgrade failed.
Return Value	Success: <code>NWY_SUCCESS</code>

Failure: other error codes

3.3 nwy_fota_base_ver_update

Format	nwy_error_e nwy_fota_base_ver_update(nwy_ota_download_param_t* param);
Description	Initiates a base version FOTA upgrade by setting download parameters. The base version will automatically download and upgrade.
Parameter	param: upgrade parameters
Return Value	Success: NWY_SUCCESS Failure: other error code
Remarks	This interface is specific to N706B and supports upgrading only the base version or both the base version and the combined APP package. Not supported in other projects.